Drift consistency of the JOREK `model600` boundary conditions: an audit and an implementation plan

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Abstract

Switching on the weak Galerkin sheath j–V boundary condition destroys the inner-leg density enhancement, with shock-capturing diffusion excluded as the cause. This document audits every boundary condition and fluid–kinetic coupling site in `model600` for consistency with the $\boldsymbol{E} \times \boldsymbol{B}$ drift, against the SOLPS-ITER manual 3.0.9.

The result: of the five boundary conditions SOLPS requires to be used together for a drift case, JOREK has one right, one half right, and three with no drift at all. The one that is half right — the Mach-1 condition — contains a *double unit conversion* rather than a missing factor, and it is *identically dormant* under the Dirichlet u boundary condition and *activates* exactly when the sheath BC is switched on. That is the same switch that produces the observed effect, which is the argument for fixing it first. Section 7 gives the implementation plan; Section 8 the case for the priority ordering; Section 9 the traps.

1 The unit convention, confirmed five ways

Everything below rests on one fact, so it is worth establishing beyond doubt:

$$\boxed{V_{\mathrm{par}} = v_\parallel / |\boldsymbol{B}|} \tag{1}$$

location	expression	implies
`mod_boundary_conditions.f90:617`	`direction/Btot * cs0`	physical speed $\div|B|$
`mod_expression.f90:1680`	`fact_vpar = sqrt(BB2)/fact_time`	$v_\parallel = V_{\mathrm{par}}|B|$
`mod_expression.f90:1487`	`Vsound = c_s/sqrt(BB2)`	same, inverted
`mod_particle_wall_interaction.f90:1770`	`vpar * norm2(B)`	explicit conversion
`mod_fields.f90`, `calc_vvector`	`v_phi = F0*vpar/R`	consistent

2 The Mach-1 bug is a double conversion, not a missing factor

The drift-compatible Bohm condition requires a correction $-\boldsymbol{v}_E \cdot \hat{\boldsymbol{n}} / b_n$ to $v_\parallel$, which in V_{par} units is $-\boldsymbol{v}_E \cdot \hat{\boldsymbol{n}} / (b_n B_{\mathrm{tot}})$. The geometric identity (derivation in `doc/jorek_vpar_bc_drifts.pdf`) gives

$$R^2 \frac{\partial_b u}{\partial_b \psi} \;=\; -\frac{\boldsymbol{v}_E \cdot \hat{\boldsymbol{n}}}{b_n\, B_{\mathrm{tot}}}, \tag{2}$$

with ∂_b the *tangential* derivative. The two sides are the same expression: $R^2 \partial_b u / \partial_b \psi$ **is already in** V_{par} **units**, the B_{tot} being generated by the $\partial_b \psi$ in the denominator. Line 617 divides it again:

```
Mach1BC = - Vpar0 + direction/Btot * factor * cs0 + factor/Btot * BigR**2 * U0_b/ps0_b
!                                                   ^^^^^ converts an already-converted quantity
```

> The `cs0` term genuinely needs $/B_{\mathrm{tot}}$, because `cs0` is a physical speed. The drift term does not, because it is not. Framing this as a *double conversion* rather than a *missing factor* matters: the former is visible in the source and verifiable against the identity, the latter invites the reasonable objection that the factor might be absorbed elsewhere.

Checked and excluded: it is not absorbed by the imposition. `boundary_conditions_add_one_entry` places $-$`zbig`$\cdot$`Mach1BC_v` on the diagonal and $+$`zbig`$\cdot$`Mach1BC` on the right-hand side, so `zbig` cancels. It is imposed on `index(1)` (the value) and `index(iv_dir)` (the *tangential* derivative). No perpendicular degree of freedom, no projection, no mass-matrix normalisation — nothing that could carry a field factor.

3 SOLPS confirms the target form

Manual 3.0.9, p. 413, `BCMOM=13`, titled *"Drift-compatible, Bohm-Chodura sheath entrance condition"*:

> *This condition imposes that the sum of the poloidal projection of the parallel velocity and the poloidal $E \times B$ velocity (restricted to not exceed $\pm 2c_{s,a}|b_x|$), equals at least the specific species sound speed $c_{s,a}$, reprojected in the poloidal direction.*

$$v_\parallel b_x + V_{\text{pol}}^{(E\times B)} \;\geq\; c_{s,a} b_x \qquad \Longrightarrow \qquad v_\parallel \geq c_{s,a} - \frac{V_{\text{pol}}^{(E\times B)}}{b_x} \tag{3}$$

At a target the poloidal direction *is* the wall normal, so $b_x \equiv b_n$ and $V_{\text{pol}}^{(E\times B)} \equiv \boldsymbol{v}_E \cdot \hat{\boldsymbol{n}}$. The $\boldsymbol{E} \times \boldsymbol{B}$ enters with coefficient **1**: `b2mn.dat` exposes `'b2stbc_cbc' '1.0'`, a multiplier on the $E \times B$ velocity in the `BCMOM=13` case. There is no $1/B$ anywhere, and the $\pm 2c_s|b_x|$ clip is theirs.

4 The audit

variable	boundary term	drift?
`u`	sheath j–V characteristic (`:605`)	**yes** (this *is* the potential BC)
`vpar`	nodal `Mach1BC` (`mod_boundary_conditions:617`)	**half** (double-converted)
`rho`	`density_reflection*r0*vpar0*ps0_s` (`:661`)	**no**
`Ti,Te,T`	$(\gamma_{\text{sheath}}-1)\cdot r_0 T$$\cdot$`vpar0`$\cdot$`ps0_s` (`:666/670/673`)	**no**
`vpar`	integral-form Mach-1 (`:679`)	**no** (inactive by default)

In SOLPS's terms: `BCPOT=11` correct, `BCMOM=13` half, `BCCON=14` and `BCENE/BCENI=15` absent — and the manual states these five must be used *in conjunction*.

4.1 Why the missing fluxes are not a weak-form pairing error

A boundary term is normally the surface term of a volume operator integrated by parts, so a missing one can indicate a broken pairing. Here it does not. The $\boldsymbol{E} \times \boldsymbol{B}$ convection of density is in *strong* form,

```
mod_elt_matrix_fft.f90:1670   + v * BigR**2 * ( r0_s * u0_t - r0_t * u0_s ) * tstep * factor(var_rho,2)
```

— test function times the bracket, *not* its derivative. It was never integrated by parts, produces no surface term, and none is missing for consistency. The code is internally consistent; the drift boundary condition was simply never written.

5 The particle module: charge exchange is right, collisions are not

Two different background velocities are used for two processes in the same particle push.

process	background velocity	status
charge exchange (`:423`$\rightarrow$`:482`)	`calc_vvector` $\rightarrow$ full $\boldsymbol{v}_E + v_\parallel \hat{\boldsymbol{b}}$	**correct**
Coulomb collisions (`:607`$\rightarrow$`:614`)	`P(1)*B/sim%t_norm` $\rightarrow$ parallel only	**bug**

calc_vvector builds v_R = -R*U_Z + vpar*psi_Z/R etc., i.e. it contains $\boldsymbol{v}_E = R(-\partial_Z u, \partial_R u)$ explicitly.

$|\boldsymbol{v}_E| \approx 170\,\mathrm{m/s}$ against $v_\parallel \sim c_s \sim 3\times 10^4\,\mathrm{m/s}$ looks like a 0.6% correction. It is not: $\boldsymbol{v}_E \perp \boldsymbol{B}$ while $v_\parallel \parallel \boldsymbol{B}$, so omitting it discards **100% of the perpendicular background flow**. Kinetic impurity ions are never collisionally entrained by the $\boldsymbol{E}\times\boldsymbol{B}$, and mom_par_source is computed against the wrong reference frame.

6 Fluid and kinetic recycling mirror each other

fluid, mod_boundary_matrix_open.f90:661	density_reflection * r0 * vpar0 * ps0_s
kinetic, mod_particle_wall_interaction.f90:1770	Gamma_d = n_e*|vpar|*norm2(B)*cos_alpha + n_e*c_s*c_angle

with cos_alpha $= |\hat{\boldsymbol{b}}\cdot\hat{\boldsymbol{n}}| = |b_n|$. Structurally identical, both parallel-only.

These must be changed together. Fixing one alone breaks particle conservation across the fluid–kinetic interface: the fluid would recycle a flux the kinetic side never creates, or the reverse.

The kinetic side is the easier of the two: the OpenMP private list at :1738 already carries U and E, and calc_RK4 returns E, so $\boldsymbol{v}_E = \boldsymbol{E}\times\boldsymbol{B}/B^2$ is directly available with no new field machinery.

7 Implementation plan

0. **Collision background flow → vvector.** particles/mod_particle_evolution.f90:614. Replace P(1)*B/sim%t_norm with vvector, already computed at :423 in the same iteration and already in $\mathrm{m\,s^{-1}}$; the interp_PRZ([var_Vpar]) at :607 then becomes redundant. Confirm vvector is computed unconditionally rather than only inside the CX branch. *Self-contained; no pairing. Makes collisions agree with charge exchange, which is already correct.*

1. **Mach1BC: remove the spurious $/B_{\mathrm{tot}}$, add the clip.** models/model600/mod_boundary_conditions.f90, behind bohm_drift_compatible = .false.:

```
:617   + factor * BigR**2 * U0_b/ps0_b            ! was factor/Btot * ...
:620   + factor * BigR**2 * element_size_0/ps0_b   ! Jacobian, same scaling
:655   + factor * BigR**2 * U0_bb/ps0_b            ! n_order >= 5 only
:656   + factor * BigR**2 * element_size_3/ps0_b   ! n_order >= 5 only
```

   and clip the drift term to ± 2 cs0. *No pairing needed: the particle module reads vpar through calc_NeTevpar and inherits the corrected value. Leave model401/502 untouched — the flag defaults off.* Self-check: the diag_mach1 ratio must go to $-\,\mathrm{sign}(b_n)$ when the flag is on.

2+3. **Recycling on the total flux, fluid *and* kinetic together.** Add $n\,(\boldsymbol{v}_E\cdot\hat{\boldsymbol{n}})$ to :661 and to Gamma_d at :1770.

4. **Sheath heat transmission on the total flux.** :666, :670, :673: apply $(\gamma_{\mathrm{sheath}}-1)$ to Γ_{tot}, not $\Gamma_\parallel$. Check the kinetic energy accounting (i_wall_heat_in) stays matched. *This is the term that couples the density asymmetry back into T_e, hence η, hence $E_\parallel$ — the amplifier of the SOLPS loop, which the boundary currently lacks entirely.*

5. **Diamagnetic velocity in the sheath condition.** SOLPS carries $u_a^{(\mathrm{dia})} b_z$ in BCMOM=3; JOREK carries it nowhere. Largest change, lowest priority.

8 Why item 1 is the most pressing

Under `bcs(:)%dirichlet%u = .true.` (the default), u is pinned along the wall, so $\partial_b u = 0$ and therefore, by (2),

the `Mach1BC` drift term is **identically zero**.

It contributes nothing, and the condition reduces to the plain Bohm condition $v_\parallel = c_s$ — which is correct, because with $\partial_b u = 0$ there is also no $\boldsymbol{E} \times \boldsymbol{B}$ flux through the wall to correct for.

Release u with the sheath BC and *both* switch on together: the drift flux becomes non-zero and the correction meant to compensate it activates — **at half size**.

The bug's activation switch is exactly the switch that produces the observed effect.

And the error is asymmetric in the right direction to explain it. Writing the total normal flux with the undersized correction,

$$\boldsymbol{v} \cdot \hat{\boldsymbol{n}} = v_\parallel b_n + \boldsymbol{v}_E \cdot \hat{\boldsymbol{n}} = c_s b_n + \underbrace{\left(1 - \frac{1}{B_{\mathrm{tot}}}\right)}_{\approx 0.5} \boldsymbol{v}_E \cdot \hat{\boldsymbol{n}}, \tag{4}$$

whereas the drift-compatible condition should give exactly $c_s b_n$. Half the drift flux survives as excess, with the sign measured by the `diag_mach1` diagnostic:

target	$\boldsymbol{v}_E \cdot \hat{\boldsymbol{n}}$	correction should	result of halving it
INNER	$+8.2 \times 10^{-7}$ (into wall)	*reduce* $v_\parallel$	too high → over-removes → **drain**
OUTER	-1.4×10^{-6} (out of wall)	*raise* $v_\parallel$	too low → under-removes → **accumulate**

It drains the high-field side and sources the low-field side — the exact direction needed to remove an inner-leg density lobe, appearing exactly when the sheath BC is enabled.

Items 2–3 push the *same* way and therefore compound rather than cancel: recycling is computed from $\Gamma_\parallel$, but under a correct drift-compatible condition $v_\parallel b_n = c_s b_n - \boldsymbol{v}_E \cdot \hat{\boldsymbol{n}}$, so at the inner target ($\boldsymbol{v}_E \cdot \hat{\boldsymbol{n}} > 0$) the parallel part is smaller and the inner target is *under*-recycled. Too much removed and too little returned, both on the HFS.

Magnitudes, stated honestly

In the mean the excess is $\sim 0.14\%$ of the Bohm flux — small. Near tangency the diagnostic measured the required correction reaching 280% of c_s, giving $\sim 140\%$ excess locally, and it is resolvable on 24% of the inner target's boundary weight against 2.3% of the outer's. So: small on average, large where it bites, and geometrically biased toward the inner target.

This is a correct fix with the right sign in the right place, and it is *not* a promise that it explains the whole lobe. The dominant effect may still be volume redistribution by the sheath-driven $\boldsymbol{E} \times \boldsymbol{B}$, which is real physics given that potential. What items 0–4 do is make the boundary consequences of the potential correct, so that the next observation can be trusted.

9 Traps

- **`c_angle` may already be standing in for the drift.** It is a minimum-flux floor at grazing incidence, and the drift correction is *largest* at grazing incidence because it goes as $1/b_n$. Adding the drift term without revisiting `c_angle` risks double counting precisely where both are biggest.

- `iv_dir` **orientation.** The fluid-side flux additions need $\partial_b u$ with the correct `normal_sign3` and `iv_dir` sense. Getting it wrong makes the term dissipative on some boundary elements and anti-dissipative on the rest — the same failure mode documented for `sheath_wall_vel` at `mod_boundary_conditions.f90:1338`, which grows a mesh-scale alternating structure in ψ, visible first in zj and only later in u.
- **Never change fluid recycling without the kinetic side**, or particle conservation breaks across the interface.
- **The clip is not optional.** The diagnostic measured a 0.89% would-reverse fraction; without the $\pm 2c_s$ bound the corrected condition can reverse $v_{||}$ near tangency.

Branch `thermoelectric-ohm`, AUG #38773, `model600`, $n_{\mathrm{tor}} = 1$. Reference: SOLPS-ITER manual 3.0.9 (`BCMOM`, pp. 407–423; `b2mn.dat` numerics). Companions: `doc/jorek_vpar_bc_drifts.pdf` (identity (2) and the Mach-1 BC in full), `doc/sheath_bc_whitepaper.pdf` (the weak Galerkin j–V BC), `doc/hfshd_findings.pdf` (campaign results).